NCL Spring 2025 Team Game Scouting Report

Dear Nic Jones (Team "SANS.edu Sentinels 391"),

Thank you for participating in the National Cyber League (NCL) Spring 2025 Season! Our goal is to prepare the next generation of cybersecurity professionals, and your participation is helping achieve that goal.

The NCL was founded in May 2011 to provide an ongoing virtual training ground for collegiate students to develop, practice, and validate their cybersecurity skills in preparation for further learning, industry certifications, and career readiness. The NCL scenario-based challenges were designed around performance-based exam objectives of CompTIA certifications and are aligned to the National Initiative for Cybersecurity Education (NICE) Cybersecurity Workforce Framework published by the National Institute of Standards and Technology (NIST).

As you look to a future career in cybersecurity, we hope you find this report to be valuable in both validating skills and identifying areas for improvement across the nine NCL skills categories. You can use this NCL Scouting Report to:

- Validate your skills to employers in any job application or professional portfolio;
- Show case your achievements and strengths by including the Score Card view of your performance as part of your résumé or simply sharing the validation link so that others may view the detailed version of this report.

The NCL Spring 2025 Season had 9,216 students/players and 596 faculty/coaches from 510 two- and four-year schools & 288 high schools across all 50 U.S. states registered to play. The Individual Game Capture the Flag (CTF) event took place from April 11 through April 13. The Team Game CTF event took place from April 25 through April 27. The games were conducted in real-time for students across the country. You were in the Experienced Students Bracket, consisting of students enrolled in advanced degrees or hold extensive industry working experience.

NCL is powered by Cyber Skyline's cloud-based skills evaluation platform. Cyber Skyline hosted the scenario-driven cybersecurity challenges for players to compete and track their progress in real-time.

To validate this report, please access: cyberskyline.com/report/VCE4V9JMNG4V

Congratulations for your participation in the NCL Spring 2025 Team Game! We hope you will continue to develop your knowledge and skills and make meaningful contributions as part of the Information Security workforce!

Dr. David Zeichick
NCL Commissioner

POWERED BY CYBER SKYLINE

NATIONAL CYBER LEAGUE SCORE CARD

NCL SPRING 2025 TEAM GAME

EXPERIENCED STUDENTS RANK

4TH PLACE OUT OF 457

PERCENTILE

100TH

YOUR TOP CATEGORIES

ENUMERATION & EXPLOITATION	WEB APPLICATION EXPLOITATION	CRYPTOGRAPHY
99TH PERCENTILE	99TH PERCENTILE	99TH PERCENTILE

97.6%
ACCURACY

Average: 66.4%

cyberskyline.com/report
ID: VCE4V9JMNG4V

NCL Spring 2025 Team Game

The NCL Team Game is designed for student players nationwide to compete in realtime in the categories listed below. The Team Game promotes camaraderie and evaluates the collective technical cybersecurity skills of the team members.

4 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

3000 POINTS OUT OF 3000
PERFORMANCE SCORE

97.6% ACCURACY

100.0% COMPLETION

100 th Experienced Students Percentile

Average: 1594.9 Points

Average: 66.4%

Average: 62.4%

Cryptography

325 POINTS OUT OF 325

89.5% ACCURACY

COMPLETION: 100.0%

Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.

Enumeration & Exploitation

380 POINTS OUT OF 380

100.0% ACCURACY

COMPLETION: 100.0%

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.

Forensics

300 POINTS OUT OF 300

100.0% ACCURACY

COMPLETION: 100.0%

Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.

Log Analysis

315 POINTS OUT OF 315

100.0% ACCURACY

COMPLETION: 100.0%

Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.

Network Traffic Analysis

300 POINTS OUT OF 300

100.0% ACCURACY

COMPLETION: 100.0%

Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.

Open Source Intelligence

350 POINTS OUT OF 350

96.4% ACCURACY

COMPLETION: 100.0%

Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.

Password Cracking

320 POINTS OUT OF 320

100.0% ACCURACY

COMPLETION: 100.0%

Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.

Scanning & Reconnaissance

300 POINTS OUT OF 300

100.0% ACCURACY

COMPLETION: 100.0%

Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.

Web Application Exploitation

310 POINTS OUT OF 310

100.0% ACCURACY

COMPLETION: 100.0%

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.

Note: Survey module (100 points) was excluded from this report.

Cryptography Module

Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.

6 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

99th Experienced Students Percentile

325 POINTS OUT OF 325
PERFORMANCE SCORE

Average: 155.0 Points

89.5%
ACCURACY

Average: 67.3%

100.0%
COMPLETION

Average: 59.0%

TOP NICE WORKROLES
Security Control Assessor
Secure Software Assessor
Exploitation Analyst
Cyber Operator
Security Architect

So Basic (Easy)

45 POINTS OUT OF 45 | 100.0% ACCURACY | COMPLETION: 100.0%

Decode the plaintext of messages encoded with common number bases

For Fun (Easy)

45 POINTS OUT OF 45 | 100.0% ACCURACY | COMPLETION: 100.0%

Decrypt messages that are encrypted using different shift ciphers

Crypto Therapy (Medium)

50 POINTS OUT OF 50 | 100.0% ACCURACY | COMPLETION: 100.0%

Exploit a poor implementation of RSA to decrypt sensitive data

Skip To My Lou (Medium)

60 POINTS OUT OF 60 | 100.0% ACCURACY | COMPLETION: 100.0%

Decrypt a message that was encrypted using the Skipjack cipher

Infinite Cosmic Power (Hard)

125 POINTS OUT OF 125 | 77.8% ACCURACY | COMPLETION: 100.0%

Perform a differential power attack on a power trace to obtain an AES encryption key

Enumeration & Exploitation Module

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.

5 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

99th Experienced Students Percentile

380 POINTS OUT OF 380
PERFORMANCE SCORE

Average: 179.0 Points

100.0% ACCURACY

Average: 71.9%

100.0% COMPLETION

Average: 65.2%

TOP NICE WORKROLES
Cyber Operator
Target Developer
Exploitation Analyst
Software Developer
Systems Security Analyst

Ransomed (Easy)

100 POINTS OUT OF 100

100.0% ACCURACY

COMPLETION: 100.0%

Analyze source code to bypass an XOR-based authentication mechanism

Crawling (Medium)

150 POINTS OUT OF 150

100.0% ACCURACY

COMPLETION: 100.0%

Perform a privilege escalation attack by coercing a binary with elevated permissions into running untrusted software

ExploitExhibit (Hard)

130 POINTS OUT OF 130

100.0% ACCURACY

COMPLETION: 100.0%

Exploit a C++ program by undermining the heap memory or leaking vtable pointers

Forensics Module

Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.

7 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

99th Experienced Students Percentile

300 POINTS OUT OF 300
PERFORMANCE SCORE

Average: 188.5 Points

100.0% ACCURACY

Average: 69.7%

100.0% COMPLETION

Average: 66.3%

TOP NICE WORKROLES
Cyber Defense Forensics Analyst
Cyber Crime Investigator
Cyber Defense Incident Responder
Cyber Defense Analyst

Origin Story (Easy)

100 POINTS OUT OF 100

100.0% ACCURACY

COMPLETION: 100.0%

Investigate a Git repository to find leaked secrets

Ransom Residue (Medium)

100 POINTS OUT OF 100

100.0% ACCURACY

COMPLETION: 100.0%

Perform a forensic investigation of a computer that was compromised by ransomware by analyzing a disk image file

Archive (Hard)

100 POINTS OUT OF 100

100.0% ACCURACY

COMPLETION: 100.0%

Automate the extraction of files by matching file signatures in a raw byte stream

Log Analysis Module

Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.

7 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

99th Experienced Students Percentile

315 POINTS OUT OF 315
PERFORMANCE SCORE

Average: 206.5 Points

100.0% ACCURACY

Average: 62.0%

100.0% COMPLETION

Average: 69.2%

TOP NICE WORKROLES
Cyber Defense Analyst
Systems Security Analyst
All-Source Analyst
Cyber Defense Forensics Analyst
Data Analyst

What Remains (Easy)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Analyze event logs from a Windows 7 workstation that was compromised by EternalBlue

Compromising (Medium)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Analyze a NetFlow log to identify compromised devices and malicious behavior

Coords (Hard)

115 POINTS OUT OF 115 | 100.0% ACCURACY | COMPLETION: 100.0%

Parse a custom binary file format and perform geospatial analysis of drone flight paths

Network Traffic Analysis Module

Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.

21 ST PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

96th Experienced Students Percentile

300 POINTS OUT OF 300
PERFORMANCE SCORE

Average: 197.6 Points

100.0% ACCURACY

Average: 75.2%

100.0% COMPLETION

Average: 78.7%

TOP NICE WORKROLES
Cyber Defense Analyst
All-Source Analyst
Cyber Defense Incident Responder
Target Network Analyst
Cyber Operator

HTTP Forever (Easy)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Analyze a packet capture with HTTP traffic to identify web browsing behavior

SQL Storage Secrets (Medium)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Analyze PostgreSQL traffic to discover information about a database and the binary content inside it

Ping of Domin8tion (Hard)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Analyze a packet capture to extract an image that was exfiltrated by encoding data in ICMP packet headers

Open Source Intelligence Module

Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.

7 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

99th Experienced Students Percentile

350 POINTS OUT OF 350
PERFORMANCE SCORE

Average: 233.4 Points

96.4% ACCURACY

Average: 63.1%

100.0% COMPLETION

Average: 76.5%

TOP NICE WORKROLES
Systems Security Analyst
Target Developer
System Administrator
Research & Development Specialist
Cyber Intel Planner

Rules of Conduct (Easy)

30 POINTS OUT OF 30 | 100.0% ACCURACY | COMPLETION: 100.0%

Introductory challenge on acceptable conduct during NCL

Swap (Easy)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Utilize blockchain tools to perform an analysis of cryptocurrency transactions

Granny (Medium)

60 POINTS OUT OF 60 | 100.0% ACCURACY | COMPLETION: 100.0%

Use open source tools to find information about the subject of an image

Sla.cking (Medium)

60 POINTS OUT OF 60 | 100.0% ACCURACY | COMPLETION: 100.0%

Perform introspection on deep web forums using the Internet Archive

Gramma Brasiliensis (Hard)

100 POINTS OUT OF 100 | 83.3% ACCURACY | COMPLETION: 100.0%

Conduct an open source investigation on an individual to create a timeline of their actions and cryptocurrency transactions

Password Cracking Module

Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.

24 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

95th Experienced Students Percentile

320 POINTS OUT OF 320
PERFORMANCE SCORE

Average: 226.0 Points

100.0% ACCURACY

Average: 88.6%

100.0% COMPLETION

Average: 74.5%

TOP NICE WORKROLES
Cyber Operator
Exploitation Analyst
Systems Security Analyst
Cyber Defense Incident Responder
Cyber Crime Investigator

Loaded Hash Browns (Easy)

30 POINTS OUT OF 30 | 100.0% ACCURACY | COMPLETION: 100.0%

Generate password hashes using MD5, SHA1, and SHA256

You're So Basic (Easy)

45 POINTS OUT OF 45 | 100.0% ACCURACY | COMPLETION: 100.0%

Crack MD5 password hashes for commonly used passwords

786b63642e636f6d2f393336 (Medium)

45 POINTS OUT OF 45 | 100.0% ACCURACY | COMPLETION: 100.0%

Crack Windows NTLM password hashes that may not be found in common rainbow tables

Zip (Medium)

50 POINTS OUT OF 50 | 100.0% ACCURACY | COMPLETION: 100.0%

Crack the insecure password for a protected zip file

Crunch Time (Medium)

50 POINTS OUT OF 50 | 100.0% ACCURACY | COMPLETION: 100.0%

Create a custom wordlist to crack passwords based on known password rules

Router (Hard)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Build a custom wordlist to crack passwords by augmenting permutation rules using known password complexity requirements

Scanning & Reconnaissance Module

Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.

7 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

99th Experienced Students Percentile

300 POINTS OUT OF 300
PERFORMANCE SCORE

Average: 143.7 Points

100.0% ACCURACY

Average: 56.0%

100.0% COMPLETION

Average: 48.1%

TOP NICE WORKROLES
Vulnerability Assessment Analyst
Target Network Analyst
Cyber Operations Planner
Target Developer
Security Control Assessor

LuckyCat (Easy)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Classify malware from benign samples by creating custom Yara rules

Schrödinger's Port (Medium)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Disable a local firewall and perform a port scan on an internal network

Ephemeral (Hard)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Scan a device and perform a port knocking sequence to access a hidden service

Web Application Exploitation Module

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.

5 TH PLACE OUT OF 457
EXPERIENCED STUDENTS RANK

99th Experienced Students Percentile

310 POINTS OUT OF 310
PERFORMANCE SCORE

Average: 136.2 Points

100.0% ACCURACY

Average: 84.7%

100.0% COMPLETION

Average: 46.4%

TOP NICE WORKROLES
Cyber Operator
Software Developer
Exploitation Analyst
Systems Security Analyst
Database Administrator

Doohickey Divulgence (Easy)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Perform analysis on minified source code to decode and extract sensitive information

Pierre's Store (Medium)

100 POINTS OUT OF 100 | 100.0% ACCURACY | COMPLETION: 100.0%

Perform a SQL injection attack to leak sensitive information from a web application

Cash Rules Everything Around Me (Hard)

110 POINTS OUT OF 110 | 100.0% ACCURACY | COMPLETION: 100.0%

Perform a series of cache deception attacks against a web application